I SEMESTER	L	T	P	C
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20BS1L02 : ENGINEERING CHEMISTRY LAB				

OUTCOMES: The experiments introduce volumetric analysis: Acid-Base, complexometric, Redox, Conductometric and potentiometric titrations. Then they are exposed to a few instrumental methods of chemical analysis.

COURSE OUTCOMES:

Thus at the end of the lab course, the student is exposed and able to

CO1: Identify the concentration of given solution by different methods of chemical analysis (**K3**)

CO2: Analyze the water purity by checking hardness, DO and Acidity. (**K4**)

CO3: Estimate the Cu^{+2}, Fe^{+3}, Ca^{+2}, Mg^{+2} ions and Ascorbic acid present in given solution. (**K4**)

CO4: Identify the pour and cloud point of lubricants. **(K3)**

CO5: Understand the principles of conductometric and potentiometric titrations. (**K2**)

LIST OF EXPERIMENTS:

1. Estimation of HCl using standard Na_2CO_3 through acid-base titration.
2. Estimate the total hardness of water using standardized EDTA solution through complexometric titration.
3. Estimation of $KMnO_4$ using standard $H_2C_2O_4$ through redox titration method.
4. Estimation of Dissolved Oxygen in given water sample by Wrinkler's Method
5. Determination of Ferric (Fe^{+3}) ions using standard $K_2Cr_2O_7$ solution
6. Determination of Copper (II) using standard hypo solution.
7. Estimation of strong acid by using strong base through conductometric titration method.
8. Estimation of strong acid by using strong base through potentiometric titration method.
9. Preparation of polymer (Demo).
10. Determination of Vitamin 'C'.
11. Determination of Pour and Cloud Point of lubricating oils

Reference Books

1. A Textbook of Quantitative Analysis, Arthur J. Vogel.